```
<!doctype html>
<html lang="en" class="h-full">
 <head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>FinanceAI Tracker</title>
  <script src="https://cdn.tailwindcss.com/3.4.17"></script>
  <script src="/_sdk/element_sdk.js"></script>
  <script src="/_sdk/data_sdk.js"></script>
  <link href="https://fonts.googleapis.com/css2?
family=DM+Sans:wght@300;400;500;600;700&amp;family=Space+Mono:wght@400;700&amp;dis
play=swap" rel="stylesheet">
  <script src="https://cdn.jsdelivr.net/npm/chart.js@4.4.0/dist/chart.umd.min.js"></script>
  <style>
    * { font-family: 'DM Sans', sans-serif; }
    .mono { font-family: 'Space Mono', monospace; }

    :root {
      --bg-primary: #0a0e1a;
      --bg-secondary: #111827;
      --text-primary: #e2e8f0;
      --accent-primary: #6366f1;
      --accent-secondary: #22d3ee;
    }

    .light-mode {
      --bg-primary: #f1f5f9;
      --bg-secondary: #ffffff;
      --text-primary: #1e293b;
      --accent-primary: #4f46e5;
      --accent-secondary: #0891b2;
    }

    body {
      background: var(--bg-primary);
      color: var(--text-primary);
      transition: all 0.3s ease;
    }

    .card-glass {
      background: var(--bg-secondary);
      border: 1px solid rgba(99, 102, 241, 0.15);
      border-radius: 16px;
      backdrop-filter: blur(10px);
      transition: all 0.3s ease;
    }

    .card-glass:hover {
      border-color: rgba(99, 102, 241, 0.35);
      box-shadow: 0 8px 32px rgba(99, 102, 241, 0.1);
    }

    .gradient-text {
      background: linear-gradient(135deg, var(--accent-primary), var(--accent-secondary));
      -webkit-background-clip: text;
      -webkit-text-fill-color: transparent;
    }

    .btn-primary {
      background: linear-gradient(135deg, var(--accent-primary), #8b5cf6);
```

```
  color: white;
  border: none;
  border-radius: 12px;
  padding: 10px 24px;
  font-weight: 600;
  cursor: pointer;
  transition: all 0.3s ease;
}
.btn-primary:hover {
  transform: translateY(-2px);
  box-shadow: 0 8px 25px rgba(99, 102, 241, 0.4);
}
.btn-primary:disabled {
  opacity: 0.5;
  cursor: not-allowed;
  transform: none;
  box-shadow: none;
}

.btn-secondary {
  background: rgba(99, 102, 241, 0.1);
  color: var(--accent-primary);
  border: 1px solid rgba(99, 102, 241, 0.3);
  border-radius: 12px;
  padding: 10px 24px;
  font-weight: 600;
  cursor: pointer;
  transition: all 0.3s ease;
}
.btn-secondary:hover {
  background: rgba(99, 102, 241, 0.2);
}

.input-field {
  background: rgba(99, 102, 241, 0.05);
  border: 1px solid rgba(99, 102, 241, 0.2);
  border-radius: 12px;
  padding: 12px 16px;
  color: var(--text-primary);
  width: 100%;
  outline: none;
  transition: all 0.3s ease;
}
.input-field:focus {
  border-color: var(--accent-primary);
  box-shadow: 0 0 0 3px rgba(99, 102, 241, 0.15);
}

.sidebar-item {
  padding: 12px 20px;
  border-radius: 12px;
  cursor: pointer;
  transition: all 0.2s ease;
  display: flex;
  align-items: center;
  gap: 12px;
  color: var(--text-primary);
  opacity: 0.6;
}
.sidebar-item:hover, .sidebar-item.active {
```

```
  background: rgba(99, 102, 241, 0.12);
  opacity: 1;
}
.sidebar-item.active {
  color: var(--accent-primary);
  font-weight: 600;
}

.stat-card {
  position: relative;
  overflow: hidden;
}
.stat-card::before {
  content: '';
  position: absolute;
  top: -50%;
  right: -50%;
  width: 100%;
  height: 100%;
  background: radial-gradient(circle, rgba(99, 102, 241, 0.08) 0%, transparent 70%);
  pointer-events: none;
}

.transaction-row {
  padding: 14px 20px;
  border-radius: 12px;
  transition: all 0.2s ease;
  display: flex;
  align-items: center;
  justify-content: space-between;
}
.transaction-row:hover {
  background: rgba(99, 102, 241, 0.06);
}

.toast-container {
  position: fixed;
  top: 20px;
  right: 20px;
  z-index: 9999;
  display: flex;
  flex-direction: column;
  gap: 8px;
}
.toast {
  padding: 12px 20px;
  border-radius: 12px;
  color: white;
  font-weight: 500;
  animation: slideIn 0.3s ease, fadeOut 0.3s ease 2.7s;
  max-width: 320px;
}
.toast-success { background: linear-gradient(135deg, #10b981, #059669); }
.toast-error { background: linear-gradient(135deg, #ef4444, #dc2626); }
.toast-info { background: linear-gradient(135deg, #6366f1, #8b5cf6); }

@keyframes slideIn {
  from { transform: translateX(100%); opacity: 0; }
  to { transform: translateX(0); opacity: 1; }
}
```

```
@keyframes fadeOut {
  from { opacity: 1; }
  to { opacity: 0; }
}

.fade-in {
  animation: fadeInUp 0.5s ease;
}
@keyframes fadeInUp {
  from { opacity: 0; transform: translateY(20px); }
  to { opacity: 1; transform: translateY(0); }
}

.pulse-dot {
  width: 8px;
  height: 8px;
  border-radius: 50%;
  animation: pulse 2s infinite;
}
@keyframes pulse {
  0%, 100% { opacity: 1; }
  50% { opacity: 0.4; }
}

.ai-shimmer {
  background: linear-gradient(90deg, transparent, rgba(99, 102, 241, 0.1), transparent);
  background-size: 200% 100%;
  animation: shimmer 2s infinite;
}
@keyframes shimmer {
  0% { background-position: -200% 0; }
  100% { background-position: 200% 0; }
}

.modal-overlay {
  position: fixed;
  inset: 0;
  background: rgba(0, 0, 0, 0.6);
  backdrop-filter: blur(4px);
  z-index: 1000;
  display: flex;
  align-items: center;
  justify-content: center;
  padding: 20px;
}
.modal-content {
  background: var(--bg-secondary);
  border: 1px solid rgba(99, 102, 241, 0.2);
  border-radius: 20px;
  padding: 32px;
  width: 100%;
  max-width: 480px;
  max-height: 90%;
  overflow-y: auto;
}

.confirm-overlay {
  position: fixed;
  inset: 0;
  background: rgba(0, 0, 0, 0.5);
```

```
  backdrop-filter: blur(2px);
  z-index: 1100;
  display: flex;
  align-items: center;
  justify-content: center;
  padding: 20px;
}
.confirm-box {
  background: var(--bg-secondary);
  border: 1px solid rgba(239, 68, 68, 0.3);
  border-radius: 16px;
  padding: 28px;
  max-width: 380px;
  width: 100%;
  text-align: center;
}

.category-badge {
  padding: 4px 12px;
  border-radius: 20px;
  font-size: 12px;
  font-weight: 600;
  display: inline-flex;
  align-items: center;
  gap: 4px;
}

.progress-bar {
  height: 8px;
  border-radius: 4px;
  background: rgba(99, 102, 241, 0.1);
  overflow: hidden;
}
.progress-fill {
  height: 100%;
  border-radius: 4px;
  transition: width 0.5s ease;
}

.tab-btn {
  padding: 8px 20px;
  border-radius: 10px;
  cursor: pointer;
  font-weight: 500;
  transition: all 0.2s ease;
  border: none;
  background: transparent;
  color: var(--text-primary);
  opacity: 0.5;
}
.tab-btn.active {
  background: rgba(99, 102, 241, 0.15);
  color: var(--accent-primary);
  opacity: 1;
}

select.input-field {
  appearance: none;
```

```
      background-image: url("data:image/svg+xml,%3Csvg xmlns='http://www.w3.org/2000/svg'
width='12' height='12' fill='%236366f1' viewBox='0 0 16 16'%3E%3Cpath d='M8 11L3 6h10l-5
5z'/%3E%3C/svg%3E");
      background-repeat: no-repeat;
      background-position: right 16px center;
    }

    .scrollbar-thin::-webkit-scrollbar { width: 4px; }
    .scrollbar-thin::-webkit-scrollbar-track { background: transparent; }
    .scrollbar-thin::-webkit-scrollbar-thumb {
      background: rgba(99, 102, 241, 0.3);
      border-radius: 2px;
    }

    @media (max-width: 768px) {
      .desktop-sidebar { display: none; }
      .mobile-nav { display: flex !important; }
    }
    @media (min-width: 769px) {
      .mobile-nav { display: none !important; }
    }
  </style>
  <style>body { box-sizing: border-box; }</style>
 </head>
 <body class="h-full">
  <div id="app" class="h-full w-full flex flex-col overflow-auto scrollbar-thin"><!-- Toast Container
-->
   <div id="toast-container" class="toast-container"></div><!-- Confirm Dialog -->
   <div id="confirm-dialog" class="confirm-overlay" style="display:none;">
    <div class="confirm-box">
     <div style="font-size:40px;margin-bottom:12px;">
      ⚠️
     </div>
     <h3 id="confirm-title" style="font-size:18px;font-weight:700;margin-bottom:8px;"></h3>
     <p id="confirm-message" style="font-size:14px;opacity:0.7;margin-bottom:24px;"></p>
     <div style="display:flex;gap:12px;justify-content:center;"><button id="confirm-cancel"
class="btn-secondary" style="flex:1;">Cancel</button> <button id="confirm-ok" class="btn-
primary" style="flex:1;background:linear-gradient(135deg,#ef4444,#dc2626);">Delete</button>
     </div>
    </div>
   </div><!-- Auth Screen -->
   <div id="auth-screen" class="h-full w-full flex items-center justify-center" style="min-
height:100%;">
    <div class="card-glass fade-in" style="padding:40px;width:100%;max-
width:420px;margin:20px;">
     <div style="text-align:center;margin-bottom:32px;">
      <div style="font-size:40px;margin-bottom:12px;">
       💎
      </div>
      <h1 id="auth-app-title" class="gradient-text" style="font-size:28px;font-
weight:700;">FinanceAI</h1>
      <p id="auth-subtitle" style="opacity:0.6;margin-top:8px;font-size:14px;">Smart Financial
Tracking</p>
     </div>
     <div style="display:flex;gap:4px;margin-
bottom:28px;background:rgba(99,102,241,0.05);border-radius:12px;padding:4px;"><button
id="auth-login-tab" class="tab-btn active" style="flex:1;" onclick="switchAuthTab('login')">Sign
In</button> <button id="auth-signup-tab" class="tab-btn" style="flex:1;"
onclick="switchAuthTab('signup')">Sign Up</button>
```

```
    </div><!-- Login Form -->
    <form id="login-form" onsubmit="handleLogin(event)">
     <div style="margin-bottom:16px;"><label for="login-email" style="font-size:13px;font-weight:500;display:block;margin-bottom:6px;opacity:0.7;">Email</label> <input type="email" id="login-email" class="input-field" placeholder="you@example.com" required>
     </div>
     <div style="margin-bottom:24px;"><label for="login-password" style="font-size:13px;font-weight:500;display:block;margin-bottom:6px;opacity:0.7;">Password</label> <input type="password" id="login-password" class="input-field" placeholder="••••••••" required minlength="6">
     </div><button type="submit" class="btn-primary" style="width:100%;padding:14px;">Sign In</button>
    </form><!-- Signup Form -->
    <form id="signup-form" style="display:none;" onsubmit="handleSignup(event)">
     <div style="margin-bottom:16px;"><label for="signup-name" style="font-size:13px;font-weight:500;display:block;margin-bottom:6px;opacity:0.7;">Full Name</label> <input type="text" id="signup-name" class="input-field" placeholder="John Doe" required>
     </div>
     <div style="margin-bottom:16px;"><label for="signup-email" style="font-size:13px;font-weight:500;display:block;margin-bottom:6px;opacity:0.7;">Email</label> <input type="email" id="signup-email" class="input-field" placeholder="you@example.com" required>
     </div>
     <div style="margin-bottom:24px;"><label for="signup-password" style="font-size:13px;font-weight:500;display:block;margin-bottom:6px;opacity:0.7;">Password</label> <input type="password" id="signup-password" class="input-field" placeholder="••••••••" required minlength="6">
     </div><button type="submit" class="btn-primary" style="width:100%;padding:14px;">Create Account</button>
    </form>
    <p style="text-align:center;margin-top:20px;font-size:12px;opacity:0.4;">Demo: Use any email &amp; password (min 6 chars)</p>
   </div>
  </div><!-- Main App -->
  <div id="main-app" style="display:none;" class="h-full w-full flex"><!-- Desktop Sidebar -->
   <aside class="desktop-sidebar card-glass" style="width:260px;min-height:100%;padding:24px 16px;border-radius:0;border-right:1px solid rgba(99,102,241,0.1);flex-shrink:0;display:flex;flex-direction:column;">
    <div style="display:flex;align-items:center;gap:10px;padding:0 12px;margin-bottom:32px;"><span style="font-size:28px;">💎</span>
     <h1 id="sidebar-title" class="gradient-text" style="font-size:20px;font-weight:700;">FinanceAI</h1>
    </div>
    <nav style="flex:1;" aria-label="Main navigation">
     <div class="sidebar-item active" data-page="dashboard" onclick="navigateTo('dashboard')"><span>📊</span><span>Dashboard</span>
     </div>
     <div class="sidebar-item" data-page="transactions" onclick="navigateTo('transactions')"><span>💳</span><span>Transactions</span>
     </div>
     <div class="sidebar-item" data-page="ai-insights" onclick="navigateTo('ai-insights')"><span>🤖</span><span>AI Insights</span>
     </div>
     <div class="sidebar-item" data-page="budget" onclick="navigateTo('budget')"><span>🎯</span><span>Budget Goals</span>
     </div>
     <div class="sidebar-item" data-page="profile" onclick="navigateTo('profile')"><span>👤</span><span>Profile</span>
```

```
      </div>
     </nav>
     <div style="padding:12px;border-top:1px solid rgba(99,102,241,0.1);margin-top:16px;">
      <div style="display:flex;align-items:center;gap:10px;">
       <div id="sidebar-avatar" style="width:36px;height:36px;border-
radius:10px;background:linear-gradient(135deg,var(--accent-primary),var(--accent-
secondary));display:flex;align-items:center;justify-content:center;font-weight:700;font-
size:14px;color:white;">
        J
       </div>
       <div>
        <div id="sidebar-name" style="font-size:13px;font-weight:600;"></div>
        <div style="font-size:11px;opacity:0.5;">
         Premium
        </div>
       </div>
      </div>
     </div>
    </aside><!-- Main Content -->
    <main class="flex-1 overflow-auto scrollbar-thin" style="padding:24px;min-height:100%;"><!--
Top Bar -->
     <header style="display:flex;justify-content:space-between;align-items:center;margin-
bottom:28px;">
      <div>
       <h2 id="page-title" style="font-size:24px;font-weight:700;"></h2>
       <p id="page-subtitle" style="font-size:13px;opacity:0.5;margin-top:2px;"></p>
      </div>
      <div style="display:flex;align-items:center;gap:12px;"><button id="theme-toggle"
onclick="toggleTheme()" style="background:rgba(99,102,241,0.1);border:1px solid
rgba(99,102,241,0.2);border-radius:10px;padding:8px 12px;cursor:pointer;color:var(--text-
primary);font-size:18px;" aria-label="Toggle dark/light mode">🌙</button> <button
onclick="openAddModal()" class="btn-primary" style="display:flex;align-
items:center;gap:6px;font-size:14px;"> <span style="font-size:16px;">+</span> Add </button>
<button onclick="handleLogout()" style="background:rgba(239,68,68,0.1);border:1px solid
rgba(239,68,68,0.2);border-radius:10px;padding:8px 12px;cursor:pointer;color:#ef4444;font-
size:13px;font-weight:500;">Logout</button>
      </div>
     </header><!-- Dashboard Page -->
     <section id="page-dashboard" class="fade-in"><!-- Stats Row -->
      <div style="display:grid;grid-template-columns:repeat(auto-
fit,minmax(200px,1fr));gap:16px;margin-bottom:28px;">
       <div class="card-glass stat-card" style="padding:24px;">
        <div style="font-size:12px;opacity:0.5;font-weight:500;text-transform:uppercase;letter-
spacing:1px;">
         Total Balance
        </div>
        <div id="stat-balance" class="mono gradient-text" style="font-size:28px;font-
weight:700;margin-top:8px;">
         $0.00
        </div>
        <div style="display:flex;align-items:center;gap:4px;margin-top:8px;">
         <div class="pulse-dot" style="background:#10b981;"></div><span style="font-
size:11px;color:#10b981;">Active</span>
        </div>
       </div>
       <div class="card-glass stat-card" style="padding:24px;">
        <div style="font-size:12px;opacity:0.5;font-weight:500;text-transform:uppercase;letter-
spacing:1px;">
         Income
        </div>
```

```
      <div id="stat-income" class="mono" style="font-size:28px;font-weight:700;margin-
top:8px;color:#10b981;">
       $0.00
      </div>
      <div id="stat-income-count" style="font-size:11px;opacity:0.5;margin-top:8px;">
       0 transactions
      </div>
     </div>
     <div class="card-glass stat-card" style="padding:24px;">
      <div style="font-size:12px;opacity:0.5;font-weight:500;text-transform:uppercase;letter-
spacing:1px;">
       Expenses
      </div>
      <div id="stat-expenses" class="mono" style="font-size:28px;font-weight:700;margin-
top:8px;color:#ef4444;">
       $0.00
      </div>
      <div id="stat-expense-count" style="font-size:11px;opacity:0.5;margin-top:8px;">
       0 transactions
      </div>
     </div>
     <div class="card-glass stat-card" style="padding:24px;">
      <div style="font-size:12px;opacity:0.5;font-weight:500;text-transform:uppercase;letter-
spacing:1px;">
       Savings Rate
      </div>
      <div id="stat-savings" class="mono" style="font-size:28px;font-weight:700;margin-
top:8px;color:var(--accent-secondary);">
       0%
      </div>
      <div id="stat-savings-label" style="font-size:11px;opacity:0.5;margin-top:8px;">
       of income saved
      </div>
     </div>
    </div><!-- Charts Row -->
    <div style="display:grid;grid-template-columns:repeat(auto-
fit,minmax(300px,1fr));gap:16px;margin-bottom:28px;">
     <div class="card-glass" style="padding:24px;">
      <h3 style="font-size:16px;font-weight:600;margin-bottom:16px;">Income vs Expenses</h3>
      <canvas id="bar-chart" height="200"></canvas>
     </div>
     <div class="card-glass" style="padding:24px;">
      <h3 style="font-size:16px;font-weight:600;margin-bottom:16px;">Spending by Category</
h3>
      <canvas id="pie-chart" height="200"></canvas>
     </div>
    </div><!-- Recent Transactions -->
    <div class="card-glass" style="padding:24px;">
     <div style="display:flex;justify-content:space-between;align-items:center;margin-
bottom:16px;">
      <h3 style="font-size:16px;font-weight:600;">Recent Transactions</h3><button
onclick="navigateTo('transactions')" class="btn-secondary" style="font-size:12px;padding:6px
14px;">View All</button>
     </div>
     <div id="recent-transactions-list"></div>
     <div id="recent-empty" style="text-align:center;padding:40px 0;opacity:0.4;">
      <div style="font-size:40px;margin-bottom:8px;">
       📭
      </div>
      <div>
```

No transactions yet. Add your first one!

</section><!-- Transactions Page -->
<section id="page-transactions" style="display:none;" class="fade-in">

<button class="tab-btn active" data-filter="all" onclick="filterTransactions('all',this)">All</button> <button class="tab-btn" data-filter="income" onclick="filterTransactions('income',this)">Income</button> <button class="tab-btn" data-filter="expense" onclick="filterTransactions('expense',this)">Expenses</button>
<label for="search-trans" class="sr-only" style="position:absolute;width:1px;height:1px;overflow:hidden;">Search transactions</label>
<input type="text" id="search-trans" class="input-field" placeholder="🔍 Search transactions..." style="max-width:280px;" oninput="searchTransactions(this.value)"> <button onclick="downloadReport()" class="btn-secondary" style="font-size:13px;padding:8px 16px;margin-left:auto;">📥 Export CSV</button>

💰

Start tracking your finances

Click "+ Add" to create your first transaction

</section><!-- AI Insights Page -->
<section id="page-ai-insights" style="display:none;" class="fade-in">

<div style="width:44px;height:44px;border-radius:12px;background:linear-gradient(135deg,var(--accent-primary),var(--accent-secondary));display:flex;align-items:center;justify-content:center;font-size:22px;">
🤖

<h3 style="font-size:18px;font-weight:700;">AI Financial Advisor</h3>

Analyzing your data

📈

<h4 style="font-size:15px;font-weight:600;margin-bottom:12px;">Spending Patterns</h4>

💡

<h4 style="font-size:15px;font-weight:600;margin-bottom:12px;">Budget Suggestions</h4>

🎯

<h4 style="font-size:15px;font-weight:600;margin-bottom:12px;">Savings Prediction</h4>

🚨

<h4 style="font-size:15px;font-weight:600;margin-bottom:12px;">Smart Alerts</h4>

</section><!-- Budget Page -->
<section id="page-budget" style="display:none;" class="fade-in">

<h3 style="font-size:16px;font-weight:600;margin-bottom:16px;">Set Monthly Budget Goals</h3>
<form id="budget-form" onsubmit="saveBudgetGoal(event)" style="display:grid;grid-template-columns:1fr 1fr auto;gap:12px;align-items:end;">
<label for="budget-category" style="font-size:12px;opacity:0.6;display:block;margin-bottom:4px;">Category</label> <select id="budget-category" class="input-field"> <option value="food">🍔 Food</option> <option value="rent">🏠 Rent</option> <option value="travel">✈️ Travel</option> <option value="bills">📄 Bills</option> <option value="entertainment">🎮 Entertainment</option> <option value="shopping">🛍️ Shopping</option> <option value="health">💊 Health</option> <option value="other">📦 Other</option> </select>

<label for="budget-amount" style="font-size:12px;opacity:0.6;display:block;margin-bottom:4px;">Monthly Limit</label> <input type="number" id="budget-amount" class="input-field" placeholder="500" min="1" required>
<button type="submit" class="btn-primary" style="height:46px;">Set Goal</button>
</form>

🎯

No budget goals yet

```
      </div>
      <div style="font-size:13px;margin-top:4px;">
       Set a monthly limit for any spending category
      </div>
     </div>
    </section><!-- Profile Page -->
    <section id="page-profile" style="display:none;" class="fade-in">
     <div class="card-glass" style="padding:32px;max-width:500px;">
      <div style="display:flex;align-items:center;gap:20px;margin-bottom:28px;">
       <div id="profile-avatar" style="width:72px;height:72px;border-
radius:18px;background:linear-gradient(135deg,var(--accent-primary),var(--accent-
secondary));display:flex;align-items:center;justify-content:center;font-size:28px;font-
weight:700;color:white;"></div>
       <div>
        <h3 id="profile-name" style="font-size:22px;font-weight:700;"></h3>
        <p id="profile-email" style="font-size:13px;opacity:0.5;"></p><span
style="background:rgba(16,185,129,0.15);color:#10b981;padding:2px 10px;border-
radius:8px;font-size:11px;font-weight:600;">Premium</span>
       </div>
      </div>
      <form id="profile-form" onsubmit="saveProfile(event)">
       <div style="margin-bottom:16px;"><label for="profile-name-input" style="font-
size:13px;opacity:0.6;display:block;margin-bottom:6px;">Display Name</label> <input
type="text" id="profile-name-input" class="input-field" required>
       </div>
       <div style="margin-bottom:16px;"><label for="profile-email-input" style="font-
size:13px;opacity:0.6;display:block;margin-bottom:6px;">Email</label> <input type="email"
id="profile-email-input" class="input-field" required>
       </div>
       <div style="margin-bottom:24px;"><label for="profile-currency" style="font-
size:13px;opacity:0.6;display:block;margin-bottom:6px;">Currency</label> <select id="profile-
currency" class="input-field"> <option value="$">USD ($)</option> <option value="€">EUR (€)</
option> <option value="£">GBP (£)</option> <option value="¥">JPY (¥)</option> <option
value="₹">INR (₹)</option> </select>
       </div><button type="submit" class="btn-primary">Save Profile</button>
      </form>
     </div>
    </section>
   </main><!-- Mobile Bottom Nav -->
   <nav class="mobile-nav" aria-label="Mobile navigation"
style="position:fixed;bottom:0;left:0;right:0;background:var(--bg-secondary);border-top:1px solid
rgba(99,102,241,0.15);padding:8px 16px;display:none;justify-content:space-around;z-
index:500;"><button onclick="navigateTo('dashboard')" class="sidebar-item active" data-
page="dashboard" style="flex-direction:column;gap:2px;padding:6px 10px;font-
size:11px;border:none;background:none;cursor:pointer;"> <span>📊</span><span>Home</
span> </button> <button onclick="navigateTo('transactions')" class="sidebar-item" data-
page="transactions" style="flex-direction:column;gap:2px;padding:6px 10px;font-
size:11px;border:none;background:none;cursor:pointer;"> <span>💳</span><span>Trans.</
span> </button> <button onclick="openAddModal()" style="width:48px;height:48px;border-
radius:50%;background:linear-gradient(135deg,var(--accent-primary),#8b5cf6);color:white;font-
size:24px;border:none;cursor:pointer;margin-top:-20px;box-shadow:0 4px 20px
rgba(99,102,241,0.4);">+</button> <button onclick="navigateTo('ai-insights')" class="sidebar-
item" data-page="ai-insights" style="flex-direction:column;gap:2px;padding:6px 10px;font-
size:11px;border:none;background:none;cursor:pointer;"> <span>🤖</span><span>AI</span>
</button> <button onclick="navigateTo('budget')" class="sidebar-item" data-page="budget"
style="flex-direction:column;gap:2px;padding:6px 10px;font-
size:11px;border:none;background:none;cursor:pointer;"> <span>🎯</span><span>Budget</
span> </button>
```

```
    </nav>
   </div><!-- Add/Edit Transaction Modal -->
   <div id="add-modal" class="modal-overlay" style="display:none;">
    <div class="modal-content">
     <div style="display:flex;justify-content:space-between;align-items:center;margin-bottom:24px;">
      <h3 id="modal-title" style="font-size:20px;font-weight:700;">Add Transaction</h3><button onclick="closeAddModal()" style="background:none;border:none;cursor:pointer;font-size:24px;color:var(--text-primary);opacity:0.5;">X</button>
     </div>
     <form id="transaction-form" onsubmit="saveTransaction(event)">
      <div style="display:flex;gap:4px;margin-bottom:20px;background:rgba(99,102,241,0.05);border-radius:12px;padding:4px;"><button type="button" id="type-income" class="tab-btn" style="flex:1;" onclick="setTransactionType('income')">💰 Income</button> <button type="button" id="type-expense" class="tab-btn active" style="flex:1;" onclick="setTransactionType('expense')">💸 Expense</button>
      </div>
      <div style="margin-bottom:16px;"><label for="trans-amount" style="font-size:13px;opacity:0.6;display:block;margin-bottom:6px;">Amount</label> <input type="number" id="trans-amount" class="input-field" placeholder="0.00" step="0.01" min="0.01" required>
      </div>
      <div style="margin-bottom:16px;"><label for="trans-desc" style="font-size:13px;opacity:0.6;display:block;margin-bottom:6px;">Description</label> <input type="text" id="trans-desc" class="input-field" placeholder="What was it for?" required>
      </div>
      <div style="margin-bottom:16px;"><label for="trans-category" style="font-size:13px;opacity:0.6;display:block;margin-bottom:6px;">Category</label> <select id="trans-category" class="input-field"> <option value="salary">💼 Salary</option> <option value="freelance">💻 Freelance</option> <option value="investment">📈 Investment</option> <option value="food">🍔 Food</option> <option value="rent">🏠 Rent</option> <option value="travel">✈️ Travel</option> <option value="bills">📄 Bills</option> <option value="entertainment">🎮 Entertainment</option> <option value="shopping">🛍️ Shopping</option> <option value="health">💊 Health</option> <option value="other">📦 Other</option> </select>
      </div>
      <div style="margin-bottom:24px;"><label for="trans-date" style="font-size:13px;opacity:0.6;display:block;margin-bottom:6px;">Date</label> <input type="date" id="trans-date" class="input-field" required>
      </div><input type="hidden" id="trans-edit-id" value=""> <button type="submit" id="trans-submit-btn" class="btn-primary" style="width:100%;padding:14px;">Add Transaction</button>
     </form>
    </div>
   </div>
  </div>
  <script>
   // ===== STATE =====
   let allTransactions = [];
   let filteredTransactions = [];
   let currentFilter = 'all';
   let searchQuery = '';
   let currentPage = 'dashboard';
   let currentTransType = 'expense';
   let editingId = null;
   let isDark = true;
   let barChart = null;
```

```
let pieChart = null;
let budgetGoals = {};
let confirmCallback = null;

const userProfile = {
  name: 'User',
  email: 'user@example.com',
  currency: '$'
};

const defaultConfig = {
  app_title: 'FinanceAI',
  welcome_message: 'Smart Financial Tracking',
  currency_symbol: '$',
  background_color: '#0a0e1a',
  surface_color: '#111827',
  text_color: '#e2e8f0',
  primary_action_color: '#6366f1',
  secondary_action_color: '#22d3ee',
  font_family: 'DM Sans',
  font_size: 14
};

const categoryEmojis = {
  salary: '💼', freelance: '💻', investment: '📈',
  food: '🍔', rent: '🏠', travel: '✈️', bills: '📄',
  entertainment: '🎮', shopping: '🛍️', health: '💊', other: '📦'
};
const categoryColors = {
  salary: '#10b981', freelance: '#6366f1', investment: '#f59e0b',
  food: '#ef4444', rent: '#8b5cf6', travel: '#06b6d4', bills: '#f97316',
  entertainment: '#ec4899', shopping: '#14b8a6', health: '#84cc16', other: '#94a3b8'
};

// ===== TOAST =====
function showToast(msg, type = 'info') {
  const c = document.getElementById('toast-container');
  const t = document.createElement('div');
  t.className = `toast toast-${type}`;
  t.textContent = msg;
  c.appendChild(t);
  setTimeout(() => t.remove(), 3000);
}

// ===== CONFIRM DIALOG =====
function showConfirm(title, message) {
  return new Promise(resolve => {
    document.getElementById('confirm-title').textContent = title;
    document.getElementById('confirm-message').textContent = message;
    document.getElementById('confirm-dialog').style.display = 'flex';
    confirmCallback = resolve;
  });
}
document.getElementById('confirm-cancel').addEventListener('click', () => {
  document.getElementById('confirm-dialog').style.display = 'none';
  if (confirmCallback) confirmCallback(false);
});
document.getElementById('confirm-ok').addEventListener('click', () => {
  document.getElementById('confirm-dialog').style.display = 'none';
```

```
  if (confirmCallback) confirmCallback(true);
});

// ===== AUTH =====
function switchAuthTab(tab) {
  document.getElementById('login-form').style.display = tab === 'login' ? 'block' : 'none';
  document.getElementById('signup-form').style.display = tab === 'signup' ? 'block' : 'none';
  document.getElementById('auth-login-tab').classList.toggle('active', tab === 'login');
  document.getElementById('auth-signup-tab').classList.toggle('active', tab === 'signup');
}

function handleLogin(e) {
  e.preventDefault();
  const email = document.getElementById('login-email').value;
  userProfile.email = email;
  userProfile.name = email.split('@')[0];
  enterApp();
}

function handleSignup(e) {
  e.preventDefault();
  const name = document.getElementById('signup-name').value;
  const email = document.getElementById('signup-email').value;
  userProfile.name = name;
  userProfile.email = email;
  showToast('Account created!', 'success');
  enterApp();
}

function enterApp() {
  document.getElementById('auth-screen').style.display = 'none';
  document.getElementById('main-app').style.display = 'flex';
  updateProfileUI();
  navigateTo('dashboard');
}

function handleLogout() {
  document.getElementById('main-app').style.display = 'none';
  document.getElementById('auth-screen').style.display = 'flex';
}

function updateProfileUI() {
  const initial = (userProfile.name || 'U')[0].toUpperCase();
  document.getElementById('sidebar-avatar').textContent = initial;
  document.getElementById('sidebar-name').textContent = userProfile.name;
  document.getElementById('profile-avatar').textContent = initial;
  document.getElementById('profile-name').textContent = userProfile.name;
  document.getElementById('profile-email').textContent = userProfile.email;
  document.getElementById('profile-name-input').value = userProfile.name;
  document.getElementById('profile-email-input').value = userProfile.email;
  document.getElementById('profile-currency').value = userProfile.currency;
}

function saveProfile(e) {
  e.preventDefault();
  userProfile.name = document.getElementById('profile-name-input').value;
  userProfile.email = document.getElementById('profile-email-input').value;
  userProfile.currency = document.getElementById('profile-currency').value;
  updateProfileUI();
  renderAll();
```

```
  showToast('Profile updated!', 'success');
}

// ===== NAVIGATION =====
function navigateTo(page) {
  currentPage = page;
  ['dashboard', 'transactions', 'ai-insights', 'budget', 'profile'].forEach(p => {
    const el = document.getElementById('page-' + p);
    if (el) el.style.display = p === page ? 'block' : 'none';
  });

  document.querySelectorAll('.sidebar-item[data-page]').forEach(item => {
    item.classList.toggle('active', item.dataset.page === page);
  });

  const titles = {
    dashboard: ['Dashboard', 'Your financial overview'],
    transactions: ['Transactions', 'Manage income & expenses'],
    'ai-insights': ['AI Insights', 'Smart financial analysis'],
    budget: ['Budget Goals', 'Track spending limits'],
    profile: ['Profile', 'Account settings']
  };
  document.getElementById('page-title').textContent = titles[page]?.[0] || '';
  document.getElementById('page-subtitle').textContent = titles[page]?.[1] || '';

  if (page === 'dashboard') updateCharts();
  if (page === 'ai-insights') generateAllInsights();
  if (page === 'budget') renderBudgetGoals();
}

// ===== THEME =====
function toggleTheme() {
  isDark = !isDark;
  document.body.classList.toggle('light-mode', !isDark);
  document.getElementById('theme-toggle').textContent = isDark ? '🌙' : '☀️';
  updateCharts();
}

// ===== TRANSACTION MODAL =====
function openAddModal(editRecord = null) {
  editingId = null;
  document.getElementById('trans-edit-id').value = '';
  document.getElementById('modal-title').textContent = 'Add Transaction';
  document.getElementById('trans-submit-btn').textContent = 'Add Transaction';
  document.getElementById('trans-amount').value = '';
  document.getElementById('trans-desc').value = '';
  document.getElementById('trans-date').value = new Date().toISOString().split('T')[0];
  setTransactionType('expense');

  if (editRecord) {
    editingId = editRecord.__backendId;
    document.getElementById('trans-edit-id').value = editingId;
    document.getElementById('modal-title').textContent = 'Edit Transaction';
    document.getElementById('trans-submit-btn').textContent = 'Update';
    document.getElementById('trans-amount').value = editRecord.amount;
    document.getElementById('trans-desc').value = editRecord.description;
    document.getElementById('trans-category').value = editRecord.category;
    document.getElementById('trans-date').value = editRecord.date;
    setTransactionType(editRecord.transaction_type);
  }
```

```
  document.getElementById('add-modal').style.display = 'flex';
}

function closeAddModal() {
  document.getElementById('add-modal').style.display = 'none';
}

function setTransactionType(type) {
  currentTransType = type;
  document.getElementById('type-income').classList.toggle('active', type === 'income');
  document.getElementById('type-expense').classList.toggle('active', type === 'expense');

  const catEl = document.getElementById('trans-category');
  if (type === 'income') {
    catEl.innerHTML = '<option value="salary">💼 Salary</option><option value="freelance">
💻 Freelance</option><option value="investment">📈 Investment</option><option
value="other">📦 Other</option>';
  } else {
    catEl.innerHTML = '<option value="food">🍔 Food</option><option value="rent">🏠
Rent</option><option value="travel">✈️ Travel</option><option value="bills">📄 Bills</
option><option value="entertainment">🎮 Entertainment</option><option value="shopping">🛍️
Shopping</option><option value="health">💊 Health</option><option value="other">📦
Other</option>';
  }
}

// ===== SAVE TRANSACTION =====
async function saveTransaction(e) {
  e.preventDefault();
  const btn = document.getElementById('trans-submit-btn');
  btn.disabled = true;
  btn.textContent = 'Saving...';

  const record = {
    transaction_type: currentTransType,
    amount: parseFloat(document.getElementById('trans-amount').value),
    description: document.getElementById('trans-desc').value,
    category: document.getElementById('trans-category').value,
    date: document.getElementById('trans-date').value,
    type: currentTransType
  };

  let result;
  if (editingId) {
    const existing = allTransactions.find(t => t.__backendId === editingId);
    if (existing) {
      result = await window.dataSdk.update({ ...existing, ...record });
    }
  } else {
    if (allTransactions.length >= 999) {
      showToast('Maximum 999 transactions reached. Delete some first.', 'error');
      btn.disabled = false;
      btn.textContent = editingId ? 'Update' : 'Add Transaction';
      return;
    }
    result = await window.dataSdk.create(record);
```

```
    }

    if (result && result.isOk) {
      showToast(editingId ? 'Transaction updated!' : 'Transaction added!', 'success');
      closeAddModal();
    } else {
      showToast('Failed to save. Please try again.', 'error');
    }

    btn.disabled = false;
    btn.textContent = editingId ? 'Update' : 'Add Transaction';
  }

  // ===== DELETE TRANSACTION =====
  async function deleteTransaction(backendId) {
    const confirmed = await showConfirm('Delete Transaction', 'Are you sure you want to delete
this transaction?');
    if (!confirmed) return;

    const record = allTransactions.find(t => t.__backendId === backendId);
    if (!record) return;

    const result = await window.dataSdk.delete(record);
    if (result.isOk) {
      showToast('Transaction deleted', 'success');
    } else {
      showToast('Failed to delete. Try again.', 'error');
    }
  }

  // ===== FILTERING =====
  function filterTransactions(filter, btn) {
    currentFilter = filter;
    document.querySelectorAll('#page-transactions .tab-btn').forEach(b =>
b.classList.remove('active'));
    if (btn) btn.classList.add('active');
    applyFilters();
  }

  function searchTransactions(q) {
    searchQuery = q.toLowerCase();
    applyFilters();
  }

  function applyFilters() {
    filteredTransactions = allTransactions.filter(t => {
      const matchFilter = currentFilter === 'all' || t.transaction_type === currentFilter;
      const matchSearch = !searchQuery || t.description.toLowerCase().includes(searchQuery) ||
t.category.toLowerCase().includes(searchQuery);
      return matchFilter && matchSearch;
    });
    renderTransactionsList('all-transactions-list', filteredTransactions, true);
    document.getElementById('transactions-empty').style.display = filteredTransactions.length ?
'none' : 'block';
  }

  // ===== RENDER =====
  function renderTransactionsList(containerId, data, showActions = false) {
    const container = document.getElementById(containerId);
    const existing = new Map([...container.children].map(el => [el.dataset.id, el]));
```

```
    const c = userProfile.currency;
    const sorted = [...data].sort((a, b) => new Date(b.date) - new Date(a.date));
    const ids = new Set();

    sorted.forEach(t => {
      const id = t.__backendId;
      ids.add(id);
      const isIncome = t.transaction_type === 'income';
      const emoji = categoryEmojis[t.category] || '📦';
      const color = categoryColors[t.category] || '#94a3b8';

      if (existing.has(id)) {
        const el = existing.get(id);
        el.querySelector('.t-desc').textContent = t.description;
        el.querySelector('.t-amount').textContent = `${isIncome ? '+' : '-'}${c}${t.amount.toFixed(2)}
`;
        el.querySelector('.t-amount').style.color = isIncome ? '#10b981' : '#ef4444';
        el.querySelector('.t-cat').textContent = `${emoji} ${t.category}`;
        el.querySelector('.t-date').textContent = formatDate(t.date);
      } else {
        const row = document.createElement('div');
        row.className = 'transaction-row';
        row.dataset.id = id;
        row.innerHTML = `
          <div style="display:flex;align-items:center;gap:14px;flex:1;min-width:0;">
            <div style="width:42px;height:42px;border-radius:12px;background:${color}
20;display:flex;align-items:center;justify-content:center;font-size:18px;flex-shrink:0;">${emoji}</
div>
            <div style="min-width:0;flex:1;">
              <div class="t-desc" style="font-weight:600;font-size:14px;white-
space:nowrap;overflow:hidden;text-overflow:ellipsis;">${t.description}</div>
              <div style="display:flex;gap:8px;align-items:center;margin-top:2px;">
                <span class="t-cat category-badge" style="background:${color}15;color:${color};">$
{emoji} ${t.category}</span>
                <span class="t-date" style="font-size:11px;opacity:0.4;">${formatDate(t.date)}</
span>
              </div>
            </div>
          </div>
          <div style="display:flex;align-items:center;gap:12px;">
            <span class="t-amount mono" style="font-weight:700;font-size:15px;color:${isIncome ?
'#10b981' : '#ef4444'};">${isIncome ? '+' : '-'}${c}${t.amount.toFixed(2)}</span>
            ${showActions ? `
              <button onclick="openAddModal(allTransactions.find(t=>t.__backendId==='${id}'))"
style="background:none;border:none;cursor:pointer;font-size:16px;opacity:0.4;padding:4px;"
aria-label="Edit transaction">✏️</button>
              <button onclick="deleteTransaction('${id}')"
style="background:none;border:none;cursor:pointer;font-size:16px;opacity:0.4;padding:4px;"
aria-label="Delete transaction">🗑️</button>
            `: ''}
          </div>`;
        container.appendChild(row);
      }
    });

    existing.forEach((el, id) => { if (!ids.has(id)) el.remove(); });
  }

  function formatDate(d) {
```

```
    const date = new Date(d);
    return date.toLocaleDateString('en-US', { month: 'short', day: 'numeric' });
  }

  function renderAll() {
    const c = userProfile.currency;
    const income = allTransactions.filter(t => t.transaction_type === 'income').reduce((s, t) => s +
t.amount, 0);
    const expenses = allTransactions.filter(t => t.transaction_type === 'expense').reduce((s, t) => s
+ t.amount, 0);
    const balance = income - expenses;
    const savingsRate = income > 0 ? Math.round((balance / income) * 100) : 0;

    document.getElementById('stat-balance').textContent = `${c}${balance.toFixed(2)}`;
    document.getElementById('stat-income').textContent = `${c}${income.toFixed(2)}`;
    document.getElementById('stat-expenses').textContent = `${c}${expenses.toFixed(2)}`;
    document.getElementById('stat-savings').textContent = `${savingsRate}%`;
    document.getElementById('stat-income-count').textContent = `${allTransactions.filter(t =>
t.transaction_type === 'income').length} transactions`;
    document.getElementById('stat-expense-count').textContent = `${allTransactions.filter(t =>
t.transaction_type === 'expense').length} transactions`;

    // Recent transactions (max 5)
    const recent = [...allTransactions].sort((a, b) => new Date(b.date) - new Date(a.date)).slice(0, 5);
    renderTransactionsList('recent-transactions-list', recent, false);
    document.getElementById('recent-empty').style.display = recent.length ? 'none' : 'block';

    applyFilters();
    if (currentPage === 'dashboard') updateCharts();
    if (currentPage === 'budget') renderBudgetGoals();
  }

  // ===== CHARTS =====
  function updateCharts() {
    const textColor = isDark ? '#94a3b8' : '#64748b';
    const gridColor = isDark ? 'rgba(99,102,241,0.08)' : 'rgba(0,0,0,0.06)';

    // Bar Chart - Last 6 months
    const months = [];
    const incomeData = [];
    const expenseData = [];
    const now = new Date();
    for (let i = 5; i >= 0; i--) {
      const d = new Date(now.getFullYear(), now.getMonth() - i, 1);
      const mKey = `${d.getFullYear()}-${String(d.getMonth() + 1).padStart(2, '0')}`;
      months.push(d.toLocaleDateString('en-US', { month: 'short' }));
      incomeData.push(allTransactions.filter(t => t.transaction_type === 'income' &&
t.date.startsWith(mKey)).reduce((s, t) => s + t.amount, 0));
      expenseData.push(allTransactions.filter(t => t.transaction_type === 'expense' &&
t.date.startsWith(mKey)).reduce((s, t) => s + t.amount, 0));
    }

    if (barChart) barChart.destroy();
    const barCtx = document.getElementById('bar-chart');
    if (barCtx) {
      barChart = new Chart(barCtx, {
        type: 'bar',
        data: {
          labels: months,
          datasets: [
```

```
            { label: 'Income', data: incomeData, backgroundColor: 'rgba(16,185,129,0.7)',
borderRadius: 8, barPercentage: 0.6 },
            { label: 'Expenses', data: expenseData, backgroundColor: 'rgba(239,68,68,0.7)',
borderRadius: 8, barPercentage: 0.6 }
          ]
        },
        options: {
          responsive: true,
          maintainAspectRatio: false,
          plugins: { legend: { labels: { color: textColor, font: { family: 'DM Sans' } } } },
          scales: {
            x: { grid: { display: false }, ticks: { color: textColor, font: { family: 'DM Sans' } } },
            y: { grid: { color: gridColor }, ticks: { color: textColor, font: { family: 'DM Sans' } } }
          }
        }
      });
    }

    // Pie Chart
    const expensesByCategory = {};
    allTransactions.filter(t => t.transaction_type === 'expense').forEach(t => {
      expensesByCategory[t.category] = (expensesByCategory[t.category] || 0) + t.amount;
    });

    const pieLabels = Object.keys(expensesByCategory);
    const pieData = Object.values(expensesByCategory);
    const pieColors = pieLabels.map(c => categoryColors[c] || '#94a3b8');

    if (pieChart) pieChart.destroy();
    const pieCtx = document.getElementById('pie-chart');
    if (pieCtx) {
      pieChart = new Chart(pieCtx, {
        type: 'doughnut',
        data: {
          labels: pieLabels.map(l => `${categoryEmojis[l] || '📦'} ${l}`),
          datasets: [{ data: pieData.length ? pieData : [1], backgroundColor: pieColors.length ?
pieColors : ['#1e293b'], borderWidth: 0, spacing: 2 }]
        },
        options: {
          responsive: true,
          maintainAspectRatio: false,
          cutout: '65%',
          plugins: {
            legend: { position: 'bottom', labels: { color: textColor, font: { family: 'DM Sans', size: 11 },
padding: 12, usePointStyle: true, pointStyleWidth: 8 } }
          }
        }
      });
    }
  }

  // ===== AI INSIGHTS =====
  function generateAIInsights() {
    const c = userProfile.currency;
    const income = allTransactions.filter(t => t.transaction_type === 'income').reduce((s, t) => s +
t.amount, 0);
    const expenses = allTransactions.filter(t => t.transaction_type === 'expense').reduce((s, t) => s
+ t.amount, 0);
    const balance = income - expenses;
```

```
    const expensesByCategory = {};
    allTransactions.filter(t => t.transaction_type === 'expense').forEach(t => {
      expensesByCategory[t.category] = (expensesByCategory[t.category] || 0) + t.amount;
    });

    const topCategories = Object.entries(expensesByCategory).sort((a, b) => b[1] - a[1]);
    const topCat = topCategories[0];

    if (!allTransactions.length) {
      document.getElementById('ai-summary').innerHTML = '🤖 <strong>Welcome!</strong>
Start adding your income and expenses to receive personalized AI financial insights, budget
suggestions, and spending analysis.';
      document.getElementById('ai-spending-content').innerHTML = 'Add transactions to see your
spending patterns analyzed by our AI.';
      document.getElementById('ai-budget-content').innerHTML = 'Once you have data, I\'ll
suggest optimal budgets for each category.';
      document.getElementById('ai-savings-content').innerHTML = 'Your savings predictions will
appear here after tracking begins.';
      document.getElementById('ai-alerts-content').innerHTML = 'Smart overspending alerts will
be generated based on your patterns.';
      return;
    }

    const savingsRate = income > 0 ? ((balance / income) * 100).toFixed(1) : 0;
    const avgExpense = expenses / (allTransactions.filter(t => t.transaction_type ===
'expense').length || 1);

    // Summary
    let summaryHTML = `<strong>📊 Financial Summary:</strong> You've earned <strong
style="color:#10b981;">${c}${income.toFixed(2)}</strong> and spent <strong
style="color:#ef4444;">${c}${expenses.toFixed(2)}</strong>.`;
    if (balance > 0) {
      summaryHTML += ` Your net savings of <strong class="gradient-text">${c}$
{balance.toFixed(2)}</strong> shows ${savingsRate}% savings rate — ${savingsRate > 20 ?
'excellent discipline! 🌟' : savingsRate > 10 ? 'good progress! Keep it up. 💪' : 'room for
improvement. Let\'s optimize! 🎯'}`;
    } else {
      summaryHTML += ` <strong style="color:#ef4444;">⚠️ You're spending more than
earning.</strong> Let's work on reducing expenses.`;
    }
    document.getElementById('ai-summary').innerHTML = summaryHTML;

    // Spending Patterns
    let spendingHTML = '';
    if (topCat) {
      spendingHTML += `Your highest spending category is <strong>${categoryEmojis[topCat[0]] ||
'📦'} ${topCat[0]}</strong> at ${c}${topCat[1].toFixed(2)} (${((topCat[1] / expenses) *
100).toFixed(0)}% of expenses).<br><br>`;
    }
    spendingHTML += `Average transaction: ${c}${avgExpense.toFixed(2)}<br>`;
    spendingHTML += `Total categories tracked: ${topCategories.length}`;
    document.getElementById('ai-spending-content').innerHTML = spendingHTML;

    // Budget Suggestions
    let budgetHTML = '';
    topCategories.slice(0, 4).forEach(([cat, amt]) => {
      const suggested = Math.ceil(amt * 0.85 / 10) * 10;
```

```
      budgetHTML += `<div style="margin-bottom:8px;">${categoryEmojis[cat] || '📦'} <strong>${cat}</strong>: Suggest ${c}${suggested}/mo (save ~${c}${(amt - suggested).toFixed(0)})</div>`;
    });
    if (!budgetHTML) budgetHTML = 'Add more expense data for personalized suggestions.';
    document.getElementById('ai-budget-content').innerHTML = budgetHTML;

    // Savings Prediction
    const monthlyIncome = income;
    const monthlyExpenses = expenses;
    const projectedSavings3 = (monthlyIncome - monthlyExpenses) * 3;
    const projectedSavings12 = (monthlyIncome - monthlyExpenses) * 12;
    document.getElementById('ai-savings-content').innerHTML = `
      Based on current patterns:<br><br>
      <strong>3 months:</strong> ${c}${projectedSavings3.toFixed(2)}<br>
      <strong>12 months:</strong> ${c}${projectedSavings12.toFixed(2)}<br><br>
      ${projectedSavings12 > 0 ? '📈 On track for positive savings!' : '📉 Consider reducing expenses to save.'}
    `;

    // Smart Alerts
    let alertsHTML = '';
    const highSpend = topCategories.filter(([_, a]) => a > income * 0.3);
    if (highSpend.length) {
      highSpend.forEach(([cat, amt]) => {
        alertsHTML += `<div style="color:#f59e0b;margin-bottom:8px;">⚠️ <strong>${cat}</strong> spending (${c}${amt.toFixed(2)}) exceeds 30% of income</div>`;
      });
    }
    if (balance < 0) alertsHTML += '<div style="color:#ef4444;margin-bottom:8px;">🚨 <strong>Overspending Alert!</strong> Expenses exceed income this period</div>';
    if (savingsRate < 10 && income > 0) alertsHTML += '<div style="color:#f97316;margin-bottom:8px;">💡 Savings rate below 10%. Aim for 20%+ for financial health.</div>';
    if (!alertsHTML) alertsHTML = '<div style="color:#10b981;">✅ Looking good! No spending alerts at this time.</div>';
    document.getElementById('ai-alerts-content').innerHTML = alertsHTML;
  }

  // ===== BUDGET GOALS =====
  function saveBudgetGoal(e) {
    e.preventDefault();
    const cat = document.getElementById('budget-category').value;
    const amt = parseFloat(document.getElementById('budget-amount').value);
    budgetGoals[cat] = amt;
    renderBudgetGoals();
    showToast(`Budget set for ${cat}!`, 'success');
    document.getElementById('budget-amount').value = '';
  }

  function renderBudgetGoals() {
    const container = document.getElementById('budget-goals-list');
    const c = userProfile.currency;
    container.innerHTML = '';

    const cats = Object.keys(budgetGoals);
    document.getElementById('budget-empty').style.display = cats.length ? 'none' : 'block';

    cats.forEach(cat => {
```

```
      const limit = budgetGoals[cat];
      const spent = allTransactions.filter(t => t.transaction_type === 'expense' && t.category ===
cat).reduce((s, t) => s + t.amount, 0);
      const pct = Math.min((spent / limit) * 100, 100);
      const color = pct > 90 ? '#ef4444' : pct > 70 ? '#f59e0b' : '#10b981';

      const card = document.createElement('div');
      card.className = 'card-glass';
      card.style.padding = '24px';
      card.innerHTML = `
        <div style="display:flex;justify-content:space-between;align-items:center;margin-
bottom:12px;">
          <div style="display:flex;align-items:center;gap:8px;">
            <span style="font-size:22px;">${categoryEmojis[cat] || '📦'}</span>
            <span style="font-weight:600;text-transform:capitalize;">${cat}</span>
          </div>
          <button onclick="delete budgetGoals['${cat}'];renderBudgetGoals();"
style="background:none;border:none;cursor:pointer;opacity:0.4;font-size:14px;">X</button>
        </div>
        <div style="display:flex;justify-content:space-between;margin-bottom:8px;font-size:13px;">
          <span style="opacity:0.6;">Spent: <strong style="color:${color};">${c}${spent.toFixed(2)}
</strong></span>
          <span style="opacity:0.6;">Limit: ${c}${limit.toFixed(2)}</span>
        </div>
        <div class="progress-bar">
          <div class="progress-fill" style="width:${pct}%;background:${color};"></div>
        </div>
        <div style="text-align:right;margin-top:6px;font-size:11px;opacity:0.5;">${pct.toFixed(0)}%
used</div>
      `;
      container.appendChild(card);
    });
  }

  // ===== EXPORT CSV =====
  function downloadReport() {
    if (!allTransactions.length) {
      showToast('No data to export', 'error');
      return;
    }
    const headers = ['Date', 'Type', 'Category', 'Description', 'Amount'];
    const rows = allTransactions.map(t => [t.date, t.transaction_type, t.category, `"$
{t.description}"`, t.amount]);
    const csv = [headers.join(','), ...rows.map(r => r.join(','))].join('\n');
    const blob = new Blob([csv], { type: 'text/csv' });
    const url = URL.createObjectURL(blob);
    const a = document.createElement('a');
    a.href = url;
    a.download = `finance-report-${new Date().toISOString().split('T')[0]}.csv`;
    a.click();
    URL.revokeObjectURL(url);
    showToast('Report downloaded!', 'success');
  }

  // ===== DATA SDK =====
  const dataHandler = {
    onDataChanged(data) {
      allTransactions = data.map(d => ({
        ...d,
        amount: typeof d.amount === 'number' ? d.amount : parseFloat(d.amount) || 0
```

```
      }));
      renderAll();
    }
  };

  // ===== ELEMENT SDK =====
  function applyConfig(config) {
    const bg = config.background_color || defaultConfig.background_color;
    const surface = config.surface_color || defaultConfig.surface_color;
    const text = config.text_color || defaultConfig.text_color;
    const primary = config.primary_action_color || defaultConfig.primary_action_color;
    const secondary = config.secondary_action_color || defaultConfig.secondary_action_color;
    const font = config.font_family || defaultConfig.font_family;
    const fontSize = config.font_size || defaultConfig.font_size;

    document.documentElement.style.setProperty('--bg-primary', bg);
    document.documentElement.style.setProperty('--bg-secondary', surface);
    document.documentElement.style.setProperty('--text-primary', text);
    document.documentElement.style.setProperty('--accent-primary', primary);
    document.documentElement.style.setProperty('--accent-secondary', secondary);

    document.body.style.fontFamily = `${font}, DM Sans, sans-serif`;
    document.body.style.fontSize = `${fontSize}px`;

    // Text content
    const title = config.app_title || defaultConfig.app_title;
    const welcome = config.welcome_message || defaultConfig.welcome_message;
    const currency = config.currency_symbol || defaultConfig.currency_symbol;

    const authTitle = document.getElementById('auth-app-title');
    if (authTitle) authTitle.textContent = title;
    const authSub = document.getElementById('auth-subtitle');
    if (authSub) authSub.textContent = welcome;
    const sideTitle = document.getElementById('sidebar-title');
    if (sideTitle) sideTitle.textContent = title;

    if (currency !== userProfile.currency) {
      userProfile.currency = currency;
      renderAll();
    }

    // Scale text
    document.querySelectorAll('h1, h2').forEach(el => el.style.fontFamily = `${font}, DM Sans,
sans-serif`);
  }

  // ===== INIT =====
  (async function init() {
    // Init Element SDK
    if (window.elementSdk) {
      window.elementSdk.init({
        defaultConfig,
        onConfigChange: async (config) => {
          applyConfig(config);
        },
        mapToCapabilities: (config) => ({
          recolorables: [
            { get: () => config.background_color || defaultConfig.background_color, set: (v) =>
{ config.background_color = v; window.elementSdk.setConfig({ background_color: v }); } },
```

```
            { get: () => config.surface_color || defaultConfig.surface_color, set: (v) =>
{ config.surface_color = v; window.elementSdk.setConfig({ surface_color: v }); } },
            { get: () => config.text_color || defaultConfig.text_color, set: (v) => { config.text_color = v;
window.elementSdk.setConfig({ text_color: v }); } },
            { get: () => config.primary_action_color || defaultConfig.primary_action_color, set: (v) =>
{ config.primary_action_color = v; window.elementSdk.setConfig({ primary_action_color: v }); } },
            { get: () => config.secondary_action_color || defaultConfig.secondary_action_color, set: (v)
=> { config.secondary_action_color = v; window.elementSdk.setConfig({ secondary_action_color:
v }); } }
          ],
          borderables: [],
          fontEditable: {
            get: () => config.font_family || defaultConfig.font_family,
            set: (v) => { config.font_family = v; window.elementSdk.setConfig({ font_family: v }); }
          },
          fontSizeable: {
            get: () => config.font_size || defaultConfig.font_size,
            set: (v) => { config.font_size = v; window.elementSdk.setConfig({ font_size: v }); }
          }
        }),
        mapToEditPanelValues: (config) => new Map([
          ['app_title', config.app_title || defaultConfig.app_title],
          ['welcome_message', config.welcome_message || defaultConfig.welcome_message],
          ['currency_symbol', config.currency_symbol || defaultConfig.currency_symbol]
        ])
      });
    }

    // Init Data SDK
    if (window.dataSdk) {
      const result = await window.dataSdk.init(dataHandler);
      if (!result.isOk) {
        console.error('Data SDK init failed');
      }
    }

    // Set default date
    document.getElementById('trans-date').value = new Date().toISOString().split('T')[0];
  })();
 </script>
 <script>(function(){function c(){var b=a.contentDocument||a.contentWindow.document;if(b){var
d=b.createElement('script');d.innerHTML="window.__CF$cv$params={r:'9ce63b55c7c24700',t:'M
Tc3MTE3MzQ2Ny4wMDAwMDA='};var a=document.createElement('script');a.nonce='';a.src='/
cdn-cgi/challenge-platform/scripts/jsd/main.js';document.getElementsByTagName('head')
[0].appendChild(a);";b.getElementsByTagName('head')[0].appendChild(d)}}if(document.body){var
a=document.createElement('iframe');a.height=1;a.width=1;a.style.position='absolute';a.style.top=
0;a.style.left=0;a.style.border='none';a.style.visibility='hidden';document.body.appendChild(a);if('l
oading'!==document.readyState)c();else
if(window.addEventListener)document.addEventListener('DOMContentLoaded',c);else{var
e=document.onreadystatechange||function(){};document.onreadystatechange=function(b)
{e(b);'loading'!==document.readyState&&(document.onreadystatechange=e,c())}}}})();</script></
body>
</html>
```